Replenishment policy for California FOODS — recommendation

To: COO · From: Yaswanth Krishna · Scope: 1 DC → 4 CA stores, 1,437 FOODS items, 56 days from 28 Mar 2016 (data to 27 Mar) · All figures: 100 simulated runs; ranges are 5th–95th percentiles; source `reports/results_1885.md`

Recommendation

Replace the current base-stock rule with a **forecast-driven, two-level policy**:

- **Stores, every day:** top each item up to what it will sell in the next 4 days in 99 out of 100 cases (99.5% for FOODS_1 and FOODS_3), never above the 21-day shelf limit. When the DC can't ship everything, ship first where a missing unit costs most (price × chance of running out).
- **DC, every Monday, per department:** order enough that the DC and the stores together stay stocked until the next supplier delivery lands, **assuming that delivery may be late**. A PO that is already overdue makes the next order bigger rather than being counted as stock on the shelf.
- Forecasts come from a model that corrects for days when shelves were empty, so it does not learn that "out of stock" means "no demand".

56 days	Current rule (base-stock)	Recommended	Lean variant[1]
Expected total cost	$141.8k ($119k–$178k)	**$114.1k ($95k–$150k)**	$95.6k
Fill rate (share of demand served)	85.7% (79.4–89.7%)	**95.2% (89.5–98.0%)**	94.8%
Chance of reaching 95%	0%	**63%**	63%
Lost-sale cost	$101.9k	$29.3k	$32.3k
Inventory left at the end (retail value)	$0.44M (13 days)	$1.50M (44 days)	$1.00M (29 days)

Savings: $27.8k (−20%) while serving 9.6 points more of demand. Lost sales fall by $72k; we spend $37k more on holding stock and $8k more on DC→store shipping (we ship what we now sell).

A decision for you — how much insurance to buy. The recommended safety levels were set on the three *previous* 8-week periods (never on the period above). There they beat the lean variant by 0.8–1.4 fill points, and they are what kept the hard January–February period close to 95%. In this spring period both did about equally well on identical simulations (95.0% vs 94.8%), so the lean variant would have saved ~$19k. Recommended if 95% must hold in difficult months; lean if ~94.5–95% in those months is acceptable.

[1] Same policy with lower safety levels (97% store and DC targets, 3-day store cover); 40 runs.

Why this works

A lost sale costs 35% of the price; holding the same unit costs 0.07% per day. **One lost sale costs as much as holding the unit for 511 days**, so the right policy keeps shelves full and pays for stock. What actually limits service is physical: the DC can ship only 13,442 units a day, while weekend demand runs ~14,300; suppliers deliver weekly with long, uncertain lead times (FOODS_3: 10 days on average, 17.5 days one time in twenty).

Recorded sales understate demand. Shelves were empty on ~4.5% of item-days, hiding ~5% of demand; correcting for this matters in money: driving the same policy with a forecast trained on raw sales costs **$4.7k more and loses 1.1 fill points** (chance of 95% falls from 63% to 50%). It also matters for *measuring* any policy: replaying recorded sales would overstate the current rule's fill by 3.5 points.

Stress test: FOODS_3 lead times double for three weeks

	Current rule	Recommended	Lean variant
No warning	**−6.4 fill points, +$32.9k**	−0.1 points, −$0.9k	−0.6 points, +$1.8k
Two weeks' warning	same (rule can't use it)	+$2.2k, no fill change	−0.1 points, +$2.7k

The recommended policy already holds 30–40 days of cover at the DC, so a 10-day slip is absorbed; the lean variant loses 0.6 points without warning and recovers almost all of it with warning. **With two weeks' notice:** (1) feed the new lead time into the Monday orders for the two weeks before (the policy does this automatically and pre-buys); (2) pre-position FOODS_3 in stores on weekdays, when DC→store capacity is spare; (3) don't over-react — with the recommended settings, extra pre-buying added $2.2k for no service gain; the warning is worth most to a lean policy. The current rule has no buffer and loses 6 points of fill.

Three biggest risks

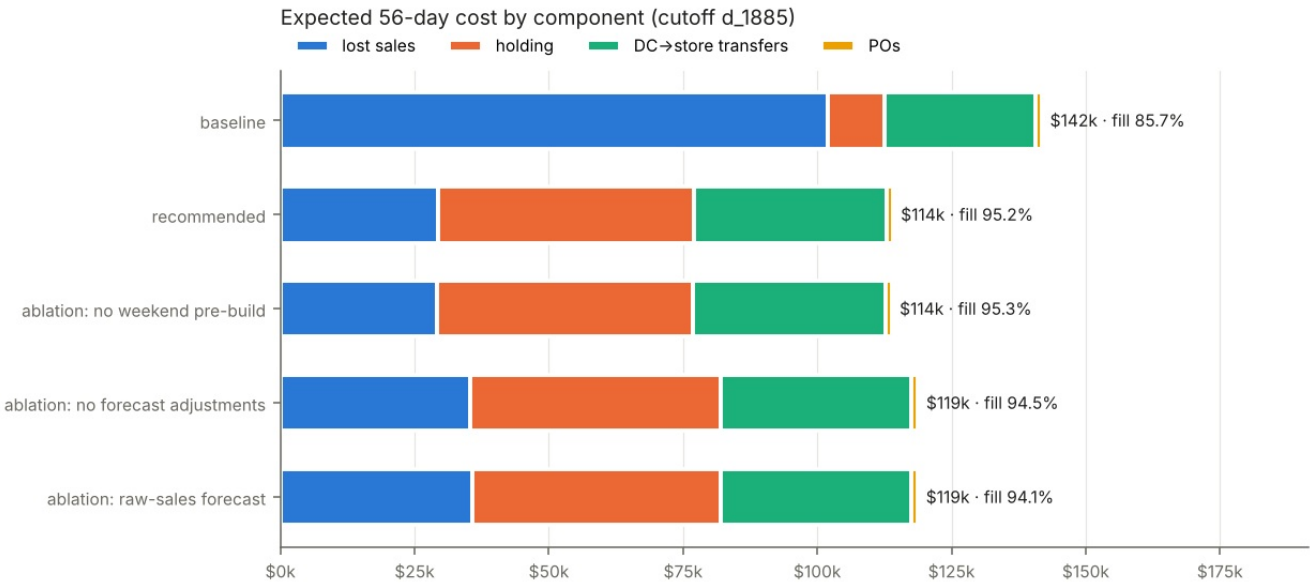
1. **A slow first delivery.** We start with 7 days of stock and nothing on order. If the first FOODS_3 PO takes

~18+ days, daily fill drops to ~70% in weeks 2–3 whatever we order. This, not the policy, is why the chance of reaching 95% is 63% rather than ~100%. *Fix: start with stock on order, or expedite the first FOODS_3 PO.*

2. **The DC outbound limit and the 21-day shelf limit.** The limit is set on recorded sales and sits below weekend demand; the shelf limit uses last year's average, which starves growing items (e.g. CA_2 doubled its volume in 2015) and items restarting after a gap — about 45% of the sales we still lose once operations settle. *Fix: review both limits against true demand.*
3. **Demand we can't see.** Shelves that empty mid-afternoon still record some sales and can't be detected; demand rebounds the model hasn't seen (January ran 9% above forecast in testing); items Walmart restocks after months off. *Mitigation: the safety levels absorb part of it; better data removes it.*

Also note: the policy ends the window holding $1.5M of stock. That stock supports sales after day 56 and is cheap to hold (included in the cost above), but it is ~$1M more than the current rule leaves (the lean variant: ~$0.5M more). If that is a constraint, use the lean variant.

Where the money goes



The recommended policy swaps lost sales for holding cost. Turning off the forecast's corrections, or training it on raw sales, each costs ~$4.7k and ~1 fill point; the weekend pre-ship rule adds nothing once stores keep 4 days of cover (left in as a safeguard).

What this analysis cannot tell you

Whether the inferred stockouts are right item by item (there is no on-hand data to check them), how lead times really vary (we used the spread you specified), and how demand reacts to promotions we cannot see.

Data we would need

Daily on-hand and out-of-stock flags per item-store (to replace the inferred stockouts); history of actual supplier lead times per PO (we assumed a fixed spread); the promotion calendar; planned launches and delistings; true DC daily shipping capacity.